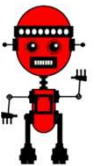
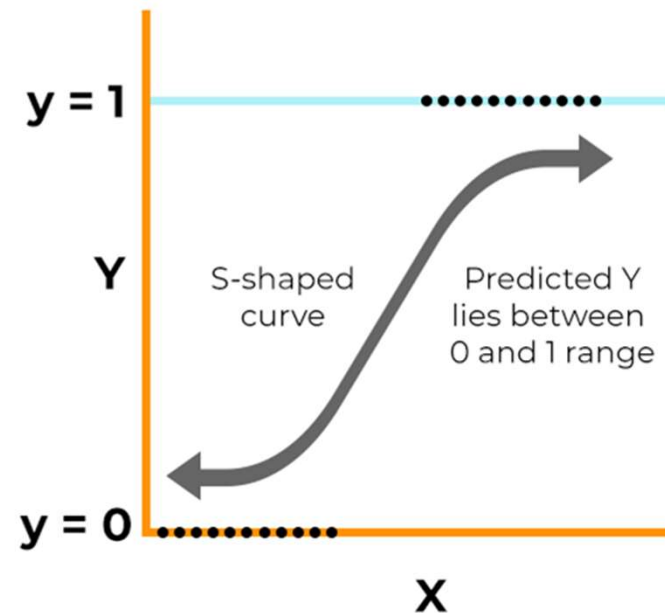


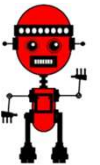
Logistic Regression Machine Learning Algorithm



CONTENT

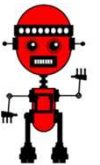
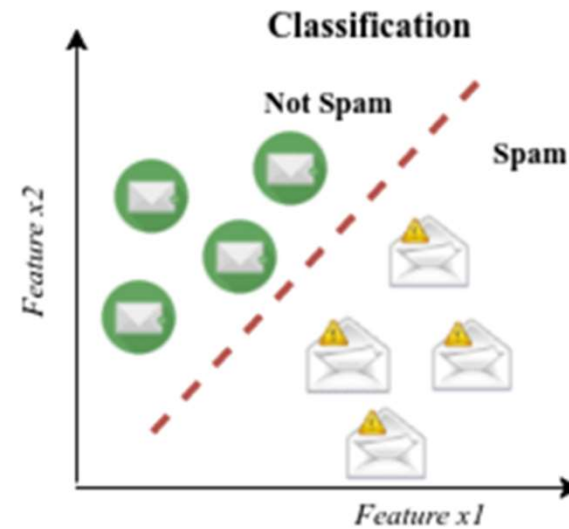
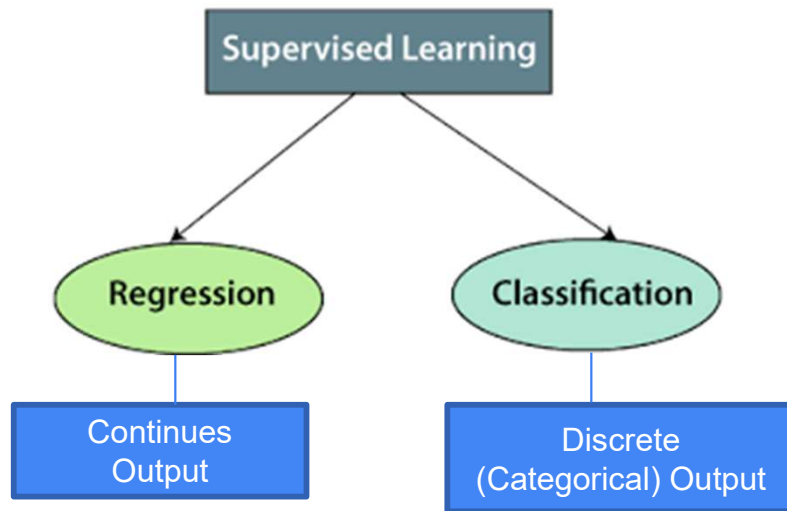


- Classification
- Types of classification
- Concept of Logistic Regression
- Logistic Function
- Use Cases
- Data Preprocessing
- Evaluation Parameters
- Take away points



What is Classification ?

Based on training data, the Classification algorithm is a Supervised Learning technique used to categorize new observations.



Types Of Classification

Binary Classification

Classification with only two class labels.

Examples comprise -

- Prediction of conversion (buy or not).
- Churn forecast (churn or not).
- Detection of spam email (spam or not).

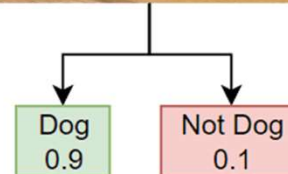
Multi-Class Classification

Classification with more than two class labels.

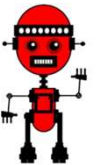
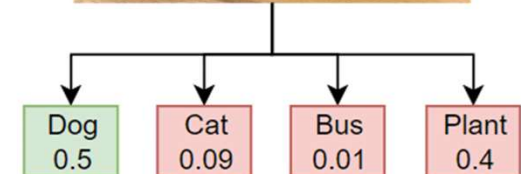
Examples comprise -

- Categorization of faces.
- Classifying plant species
- Character recognition using optical.

Binary Classification



Multiclass Classification



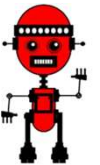
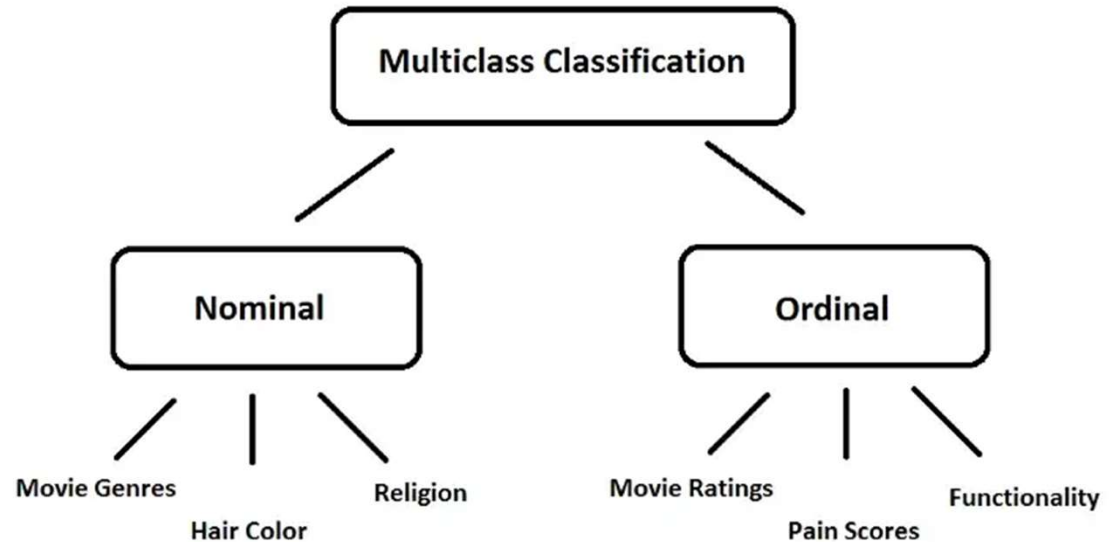
Types Of Multi - Class Classification

Nominal

- There is no apparent order.
- Examples - movie genres, hair colors, and religions.

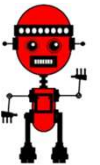
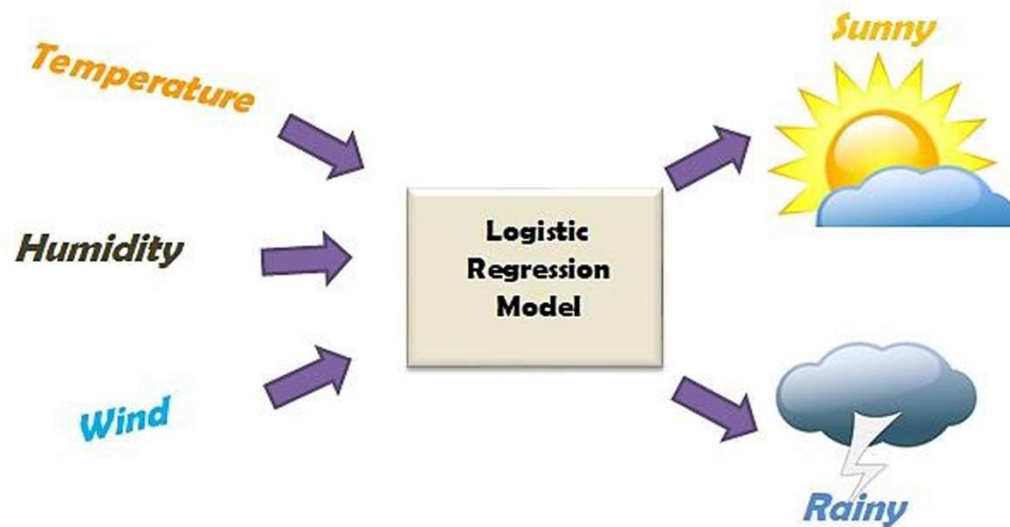
Ordinal

- There is order. “position” or “rank” among the other targets.
- Examples - movie ratings, hospital pain scores, and levels of functionality.



Logistic Regression

- Comes under supervised learning.
- Used to predict the probability of binary event occurrence(classify binary class).
- The name “logistic regression” is derived from the concept of the logistic function that it uses.

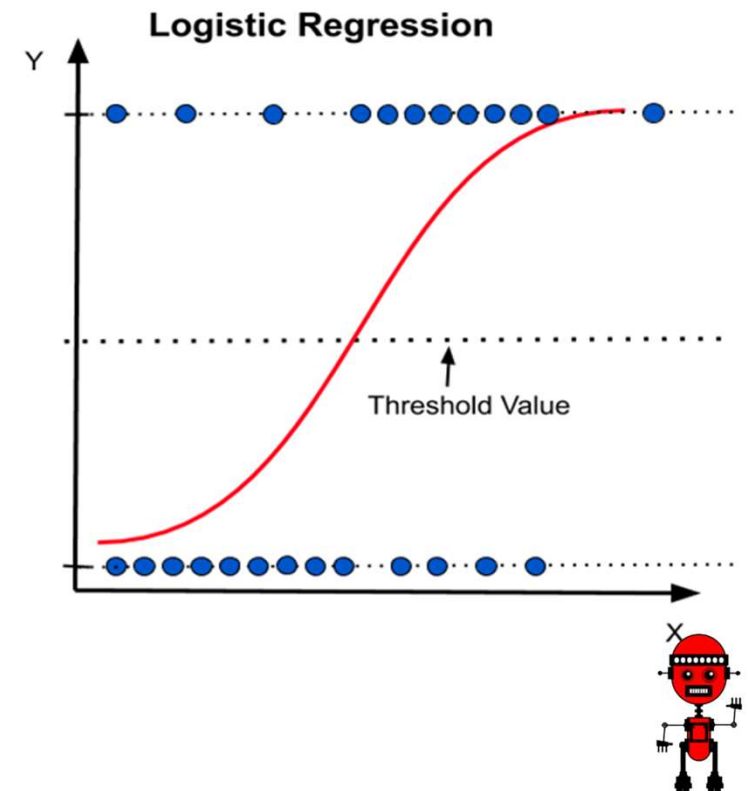


Logistic Function(Sigmoid Function)

- Used to map predicted values to probabilities between 0 and 1.
- Class is decided based on threshold value(0.5).
- Values goes above the threshold classify into Class 1 and value goes below the threshold classify into Class 2.

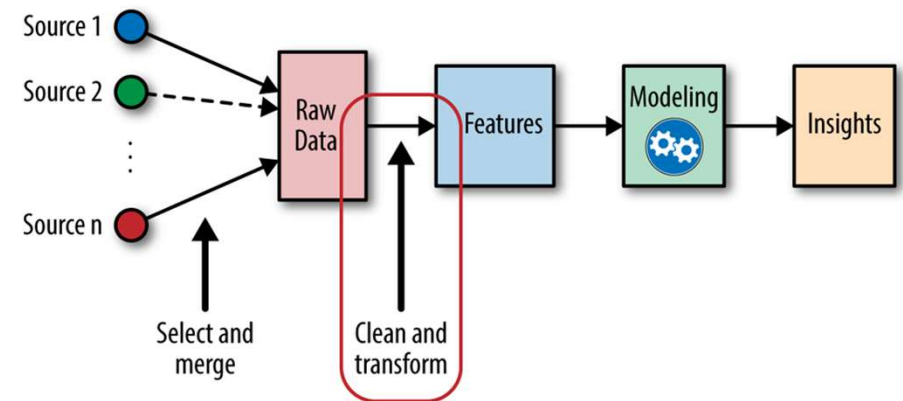
$$f(x) = \frac{1}{1 + e^{-x}}$$

- Here
 - e: Euler's constant approximately equal to 2.71828.
 - x : mx+c



Feature Engineering :

Feature engineering is the pre-processing step of machine learning, which extracts features from raw data.



Processes :

- The pre-processing step of machine learning
- Feature Creation
- Feature transformation
- Feature extraction
- Feature Selection

Techniques :

- Imputation
- Handling of categorical feature
- Handling of numerical column

Handling Categorical Features

Encoding

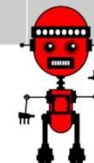
Used to transform the categorical values into numerical ones.

Types of encoding

One-hot/dummy encoding

- Converts each categorical value into a new categorical column and assign a binary value of 1 or 0 to those columns.
- Each integer value is represented as a binary vector. All the values are zero, and the index is marked with a 1.

Index	Animal	One-Hot code					
0	Dog	0	1	0	0	0	0
1	Cat	1	0	1	0	0	0
2	Sheep	2	0	0	1	0	0
3	Horse	3	0	0	0	0	1
4	Lion	4	0	0	0	1	0



Handling Categorical Features

Label encoding

- Encode target labels with value between 0 and $n_classes-1$.
- Used when the number of categories are quite large

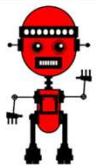
Original Data

Team	Points
A	25
A	12
B	15
B	14
B	19
B	23
C	25
C	29



Label Encoded Data

Team	Points
0	25
0	12
1	15
1	14
1	19
1	23
2	25
2	29



Handling Numerical Feature

Scaling

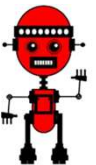
Used to normalize the range of features.

Types of scaling

Normalization / Min-Max Scaling

- Re-scales a feature value with distribution value between 0 and 1.
- Useful when there are no outliers.
- Minimum and maximum value of features are used for scaling.

$$X_{\text{new}} = \frac{X_i - \min(X)}{\max(x) - \min(X)}$$



Handling Numerical Feature

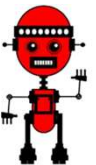
Standardization / Z - Score Normalization

- Used when we want to ensure zero mean and unit standard deviation.
- Mean and standard deviation is used for scaling.
- Much less affected by outliers.

$$z = \frac{x - \mu}{\sigma}$$

μ = Mean

σ = Standard Deviation



Confusion Matrix

A summary of prediction results on a classification problem.

Terminology

Condition positive (P)

The number of real positive cases in the data

Condition negative (N)

The number of real negative cases in the data

True positive (TP)

A test result that correctly indicates the presence of a condition or characteristic

True negative (TN)

A test result that correctly indicates the absence of a condition or characteristic

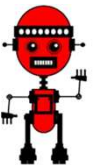
False positive (FP)

A test result which wrongly indicates that a particular condition or attribute is present

False negative (FN)

A test result which wrongly indicates that a particular condition or attribute is absent

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN



Classification Report

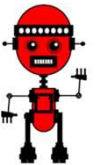
Performance evaluation matrix of your trained classification model.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

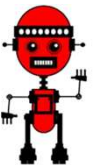
$$Recall = \frac{TP}{TP + FN}$$

$$F1-score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$



Use Cases Of Logistic Regression In Real Life

- Loan Prediction.
- Weather forecasting.
- Ecommerce companies
- Fraud detection
- Disease diagnosis



Take Away Points

- Data has no null values.
- correlation of Dependent variable('Left') with independent variable('satisfaction_level') have highest correlation with strength of -0.39.
- Used get_dummies encoding to convert categorical data(Salary,Department) into numerical.
- Used Scaling technique to standardize the independent variables in a specific range.
- Used logistic regression because dependent variable is categorical in nature.
- Used confusion matrix & classification report to determine the performance of the classification model.
- Model got training accuracy 79.4%,testing accuracy 79.6%.
- Model got precision(0=0.93,1=0.37),recall(0=0.83,1=0.61),f1 score(0=0.87,1=0.46).

